

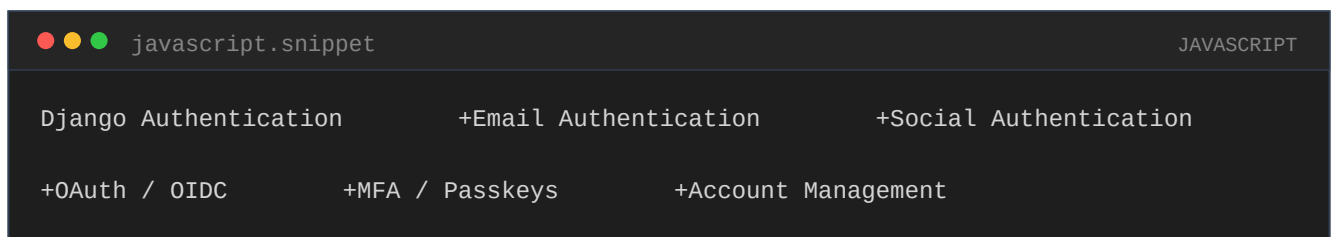
Django Allauth -- Complete Technical Notes

. Django Allauth ?

django-allauth Django- authentication framework, user authentication social authentication--

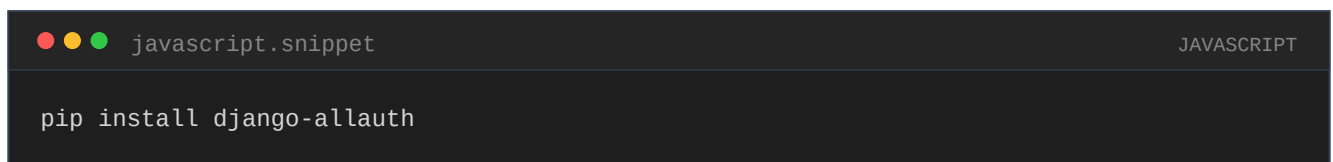
implement :

- Signup / Registration
- Login / Logout
- Email Verification
- Password Reset
- Password Change
- Social Login
- OAuth2 / OpenID Connect
- Account Management
- MFA / 2FA
- WebAuthn / Passkeys
- Headless authentication



. Installation

package install :



settings.py- apps

. Basic Configuration

settings.py

```
javascript.snippet JAVASCRIPT

INSTALLED_APPS = [    "django.contrib.admin",    "django.contrib.auth",
"django.contrib.contenttypes"
,    "django.contrib.sessions",    "django.contrib.messages",
"django.contrib.staticfiles"
,    "django.contrib.sites",    "allauth",    "allauth.account",
"allauth.socialaccount"
,    # Example provider    "allauth.socialaccount.providers.google",]SITE_ID = 1
```

Authentication Backend

```
javascript.snippet JAVASCRIPT

AUTHENTICATION_BACKENDS = [    "django.contrib.auth.backends.ModelBackend",
"allauth.account.auth_backends.AuthenticationBackend"
,]
```

django.contrib.sites :

```
javascript.snippet JAVASCRIPT

SITE_ID = 1
```

configure

. Database Migration

Installation configuration- :

```
javascript.snippet JAVASCRIPT

python manage.py migrate
```

Admin user :

```
javascript.snippet JAVASCRIPT

python manage.py createsuperuser
```

Server :

```
javascript.snippet
```

JAVASCRIPT

```
python manage.py runserver
```

. URL Configuration

urls.py:

```
javascript.snippet
```

JAVASCRIPT

```
from django.contrib import admin
from django.urls import include, path
urlpatterns = [
    path("admin/", admin.site.urls),
    path("accounts/", include("allauth.urls")),
]
```

Allauth- authentication URLs :

```
javascript.snippet
```

JAVASCRIPT

```
/accounts/login//accounts/logout//accounts/signup//accounts/password/reset//accounts/
password
/change//accounts/confirm-email/
```

. Django Allauth Configuration Settings

Allauth- settings category-

. General Settings

ACCOUNT_ADAPTER

Custom account adapter

```
javascript.snippet
```

JAVASCRIPT

```
ACCOUNT_ADAPTER = "myapp.adapters.MyAccountAdapter"
```

Adapter default behavior customize

ACCOUNT_FORMS

Built-in Allauth forms- custom form

:

```
javascript.snippet JAVASCRIPT
ACCOUNT_FORMS = {    "signup": "accounts.forms.CustomSignupForm",}
```

ACCOUNT_PREVENT_ENUMERATION

User/account existence leak

```
javascript.snippet JAVASCRIPT
ACCOUNT_PREVENT_ENUMERATION = True
```

Security-sensitive application-

ACCOUNT_RATE_LIMITS

Login, signup, password reset request rate

brute-force automated abuse

ACCOUNT_SESSION_REMEMBER

"Remember me" behavior

login session browser session- behavior

ACCOUNT_TEMPLATE_EXTENSION

Allauth template- extension

:

```
javascript.snippet JAVASCRIPT
ACCOUNT_TEMPLATE_EXTENSION = "html"
```

. Signup Settings

ACCOUNT_SIGNUP_FIELDS

Signup form- field

:

```
javascript.snippet JAVASCRIPT
ACCOUNT_SIGNUP_FIELDS = [ "email*", "username*", "password1*", "password2*"
,]
* required field
```

ACCOUNT_SIGNUP_FORM_CLASS

Signup form- custom field

```
javascript.snippet JAVASCRIPT
ACCOUNT_SIGNUP_FORM_CLASS = "accounts.forms.CustomSignupForm"
```

:

```
javascript.snippet JAVASCRIPT
UsernameEmailPasswordFirst NameLast NamePhone Number
```

ACCOUNT_SIGNUP_FORM_HONEYPOT_FIELD

Bot/spam detection- hidden honeypot field

Bot hidden field request

. Login Settings

ACCOUNT_LOGIN_METHODS

identifier login

:

```
javascript.snippet JAVASCRIPT
usernameemailusername + email
```

Configuration- exact accepted format allauth version- documentation

ACCOUNT_LOGIN_BY_CODE_ENABLED

Password- login code feature enable

```
javascript.snippet JAVASCRIPT
ACCOUNT_LOGIN_BY_CODE_ENABLED = True
```

passwordless/code-based login flow-

ACCOUNT_LOGIN_ON_EMAIL_CONFIRMATION

Email confirmation user- login

ACCOUNT_LOGIN_TIMEOUT

Login flow- timeout

login process flow expire

. Logout Settings

ACCOUNT_LOGOUT_ON_GET

GET request logout behavior

Security- logout action- POST

ACCOUNT_LOGOUT_ON_PASSWORD_CHANGE

Password session/logout behavior

. Password Management

Password Change

Authenticated user password

URL:

```
javascript.snippet JAVASCRIPT
/accounts/password/change/
```

Password Reset

Forgot password flow- :

```
javascript.snippet JAVASCRIPT
/accounts/password/reset/
```

ACCOUNT_PASSWORD_RESET_BY_CODE_ENABLED

Password reset link- code-based reset flow enable

ACCOUNT_PASSWORD_RESET_TOKEN_GENERATOR

Custom password reset token generator

Custom security requirement

. Email Verification

Email verification authentication system-

ACCOUNT_EMAIL_VERIFICATION

behavior :

```
javascript.snippet JAVASCRIPT
mandatoryoptionalnone
```

mandatory

Email verification account /

optional

User verification , verification

none

Email verification

:

```
javascript.snippet JAVASCRIPT
ACCOUNT_EMAIL_VERIFICATION = "mandatory"
```

ACCOUNT_CONFIRM_EMAIL_ON_GET

Confirmation link GET request- open confirmation

ACCOUNT_EMAIL_CONFIRMATION_EXPIRE_DAYS

Email confirmation- validity period setting

ACCOUNT_EMAIL_VERIFICATION_BY_CODE_ENABLED

Email link- verification code flow enable

. Reauthentication

Sensitive operation- user- authentication

:

```

Password changeEmail changeSecurity settings change

```

ACCOUNT_REAUTHENTICATION_REQUIRED

Sensitive action- reauthentication

ACCOUNT_REAUTHENTICATION_TIMEOUT

reauthentication valid

. Redirect Settings

Authentication flow user redirect settings

ACCOUNT_AUTHENTICATED_LOGIN_REDIRECTS

Already authenticated user login/signup page access redirect behavior

ACCOUNT_SIGNUP_REDIRECT_URL

Signup URL-

:

```

ACCOUNT_SIGNUP_REDIRECT_URL = "/dashboard/"

```

ACCOUNT_LOGOUT_REDIRECT_URL

Logout redirect destination

```

ACCOUNT_LOGOUT_REDIRECT_URL = "/"

```

ACCOUNT_EMAIL_CONFIRMATION_ANONYMOUS_REDIRECT_

Anonymous user email confirmation redirect

. Email Sending Settings

ACCOUNT_EMAIL_SUBJECT_PREFIX

Email subject- prefix

:

```
javascript.snippet JAVASCRIPT
ACCOUNT_EMAIL_SUBJECT_PREFIX = "[My Website] "
```

Email subject:

```
javascript.snippet JAVASCRIPT
[My Website] Confirm Your Email
```

ACCOUNT_EMAIL_UNKNOWN_ACCOUNTS

Unknown email address- password reset request email -- behavior

account enumeration prevention-

ACCOUNT_EMAIL_NOTIFICATIONS

Security-related email notification behavior

:

```
javascript.snippet JAVASCRIPT
Password changedEmail changedSecurity-related account event
```

. Email Address Settings

ACCOUNT_CHANGE_EMAIL

User email address email address management- behavior

ACCOUNT_EMAIL_MAX_LENGTH

Email address- maximum length

ACCOUNT_UNIQUE_EMAIL

email address account-

configuration:

```
javascript.snippet JAVASCRIPT
ACCOUNT_UNIQUE_EMAIL = True
```

. Username Settings

ACCOUNT_PRESERVE_USERNAME_CASING

Username- uppercase/lowercase formatting preserve

:

```
javascript.snippet JAVASCRIPT
FarukAhmedfarukahmed
```

ACCOUNT_USERNAME_BLACKLIST

username

:

```
adminrootadministratorsupport
```

ACCOUNT_USERNAME_MIN_LENGTH

Username- minimum length

ACCOUNT_USERNAME_VALIDATORS

Custom Django validators username validation

. Custom User Model Settings

Custom User Model Allauth- field mapping

ACCOUNT_USER_MODEL_EMAIL_FIELD

Custom User Model- email field-

ACCOUNT_USER_MODEL_USERNAME_FIELD

Custom User Model- username field-

ACCOUNT_USER_DISPLAY

User- display name customize

. Custom User Model

Django- custom user model :

```
AUTH_USER_MODEL = "accounts.User"
```

Custom user model project- configure

:

```
javascript.snippet JAVASCRIPT

class User(AbstractUser):
    pass
```

email-based authentication , custom user model- email field authentication requirements design

. Social Authentication

Allauth- feature social login

Supported provider- :

```
javascript.snippet JAVASCRIPT

GoogleGitHubFacebookAppleMicrosoftLinkedInOpenID ConnectSAML
```

Provider availability configuration package/version

. Google Login Configuration

Google provider install/configure

```
javascript.snippet JAVASCRIPT

INSTALLED_APPS = [
    # ... "allauth", "allauth.account",
    "allauth.socialaccount"
, "allauth.socialaccount.providers.google",]
```

Google OAuth credentials

. Google OAuth Flow

Authorization Code flow:

```
javascript.snippet JAVASCRIPT

User Django Application Google Authorization URL Google Login User Consent
Authorization
Code Django Callback Token Exchange User Information Django Account
```

Authorization Code Access Token

. Social Application

Django Admin Social Application configure

flow:

```
javascript.snippet JAVASCRIPT

Django Admin Social Accounts Social Applications Google Client IDClient
Secret
Sites
```

. Template Social Login

Template-provider login URL

```
javascript.snippet JAVASCRIPT

{% load socialaccount %}<a href="{% provider_login_url 'google' %}"> Login with
Google
</a>
```

. OAuth2 ?

OAuth 2.0 authorization framework

application system- resource permission- access , user- password application-

. OAuth2- Terms

Resource Owner

user, resource data

Client

application resource access

:

```
javascript.snippet JAVASCRIPT
Django ApplicationMobile AppSPA
```

Authorization Server

server authorization token issue

:

```
javascript.snippet JAVASCRIPT
GoogleMicrosoftGitHub
```

Resource Server

protected user data/resource

Access Token

Protected resource access credential

Refresh Token

access token token

. OAuth2 Authorization Code Flow

```
javascript.snippet JAVASCRIPT
User Django App Authorization Server User Login + Consent Authorization Code Django App
Token Endpoint Access Token Resource/API
```

Modern web application- Authorization Code Flow- **PKCE** , public clients-

. OAuth2 ?

OAuth2- :

- Password share authorization
- Limited permissions scopes
- Access token- expiration
- Third-party API access
- SSO architecture
- Web, SPA mobile application integrate

. JWT ?

JWT = JSON Web Token

JWT claims token format

structure:

```
javascript.snippet JAVASCRIPT
Header.Payload.Signature
:
```

```
javascript.snippet JAVASCRIPT
xxxxx.yyyyy.zzzzz
```

. JWT-Header

Token metadata

:

```
javascript.snippet JAVASCRIPT
{  "alg": "HS256",  "typ": "JWT"}
```

Payload

Claims

:

```
javascript.snippet JAVASCRIPT
{  "user_id": 123,  "role": "user"}
```

Signature

Token signature

. JWT-

JWT :

```
javascript.snippet JAVASCRIPT
REST API Mobile Application SPADistributed Services
```

JWT OAuth2

. OAuth2 JWT

	OAuth2	JWT
	Authorization framework	Token format
	Resource authorization	Claims
token format?		
Third-party authorization		
Access token	OAuth2 flow-	JWT
Refresh token	OAuth2-	JWT- requirement

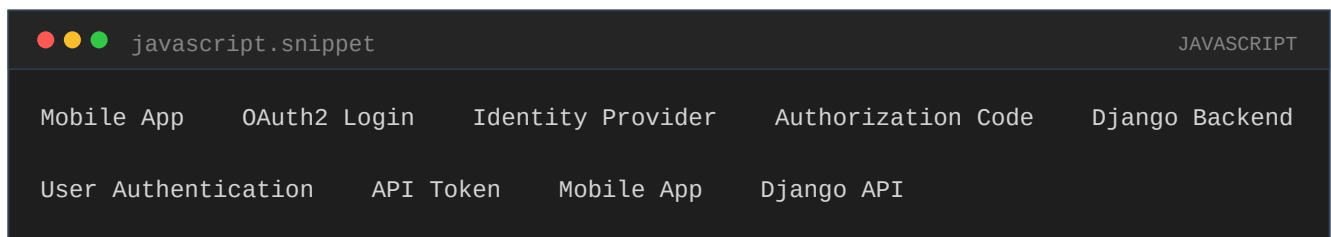
Stateless API		common
---------------	--	--------

OAuth2 JWT

OAuth2 system- access token JWT , opaque token-

. OAuth2 + JWT

architecture:



Django API- JWT

. Session Authentication JWT

	Session	JWT
State	Server-side session state	stateless
Browser		
CSRF	Cookie-based	Architecture
Revocation	Server-side	Design-dependent
Storage	Server-side session	Client token storage
API		common

JWT session- -- Application architecture authentication strategy

. Headless Allauth

Modern frontend :

```
javascript.snippet JAVASCRIPT
ReactVueNext.jsMobile App
```

Django template- API-oriented authentication architecture

Allauth- headless functionality architecture-
architecture:

```
javascript.snippet JAVASCRIPT
Frontend Django API Allauth Authentication Database
```

. MFA -- Multi-Factor Authentication

MFA authentication security

:

```
javascript.snippet JAVASCRIPT
Something you know +Something you have
```

:

```
javascript.snippet JAVASCRIPT
Password+Authenticator Code
```

. TOTP

TOTP = Time-based One-Time Password

Authenticator application code

Flow:

```
javascript.snippet JAVASCRIPT
Username + Password TOTP Code Authenticated
```

. WebAuthn / Passkeys

WebAuthn passwordless authentication

:

● ● ● javascript.snippet JAVASCRIPT

BiometricSecurity KeyDevice CredentialPaskey

:

● ● ● javascript.snippet JAVASCRIPT

FingerprintFace authenticationHardware security key

. Adapters

Allauth- default behavior Adapter

:

● ● ● javascript.snippet JAVASCRIPT

Account AdapterSocial Account Adapter

:

● ● ● javascript.snippet JAVASCRIPT

```
from allauth.account.adapter import DefaultAccountAdapter
class MyAccountAdapter(DefaultAccountAdapter):
    def get_login_redirect_url(self, request):
        return "/dashboard/"
```

:

● ● ● javascript.snippet JAVASCRIPT

ACCOUNT_ADAPTER = "accounts.adapters.MyAccountAdapter"

. SocialAccountAdapter

Social login- behavior customize

:

```
javascript.snippet JAVASCRIPT

from allauth.socialaccount.adapter import DefaultSocialAccountAdapterclass
MySocialAccountAdapter
(DefaultSocialAccountAdapter):    def save_user(self, request, sociallogin, form=None):

user = super().save_user(request, sociallogin, form)        # Custom logic
return
user
```

social login- custom business logic

. Signals

authentication event- custom action signals

:

```
javascript.snippet JAVASCRIPT

user_signed_up
```

:

```
javascript.snippet JAVASCRIPT

User signup    Signal    Welcome email    Analytics    Profile creation
```

business-critical logic- signal transaction application architecture

. Email Templates Customize

Allauth- email template override

Project- template structure :

```
javascript.snippet JAVASCRIPT

templates/ account/      email/      email_confirmation_message.txt
password_reset_key_message
.txt      ...
```

branding email customize

. Security Best Practices

Production application-:

HTTPS

```
javascript.snippet JAVASCRIPT

HTTP HTTPS
```

Strong password policy

Django password validators configure

Email verification

:

```
javascript.snippet JAVASCRIPT

ACCOUNT_EMAIL_VERIFICATION = "mandatory"
```

Rate limiting

Brute-force automated attacks

Account enumeration

```
javascript.snippet JAVASCRIPT

ACCOUNT_PREVENT_ENUMERATION = True
```

Secure cookies

Production- Django- secure cookie settings configure

Secret Git-

```
javascript.snippet JAVASCRIPT
CLIENT_SECRETSECRET_KEYAPI_KEYS
```

Environment variables secret manager

. Common Troubleshooting

: Site configuration error

:

```
javascript.snippet JAVASCRIPT
SITE_ID = 1
```

Django Admin-:

```
javascript.snippet JAVASCRIPT
Sites
```

configuration

: Social login

:

```
javascript.snippet JAVASCRIPT
Client IDClient SecretRedirect URIProvider configurationSITE_IDSocial
ApplicationSelected
Sites
```

: Migration error

:

```
javascript.snippet JAVASCRIPT

python manage.py makemigrationspython manage.py migrate
```

: Email

Development- console backend :

```
javascript.snippet JAVASCRIPT

EMAIL_BACKEND = "django.core.mail.backends.console.EmailBackend"
```

email terminal/console-

Production- SMTP transactional email service configure

. Important Configuration Example

basic email-based authentication setup- :

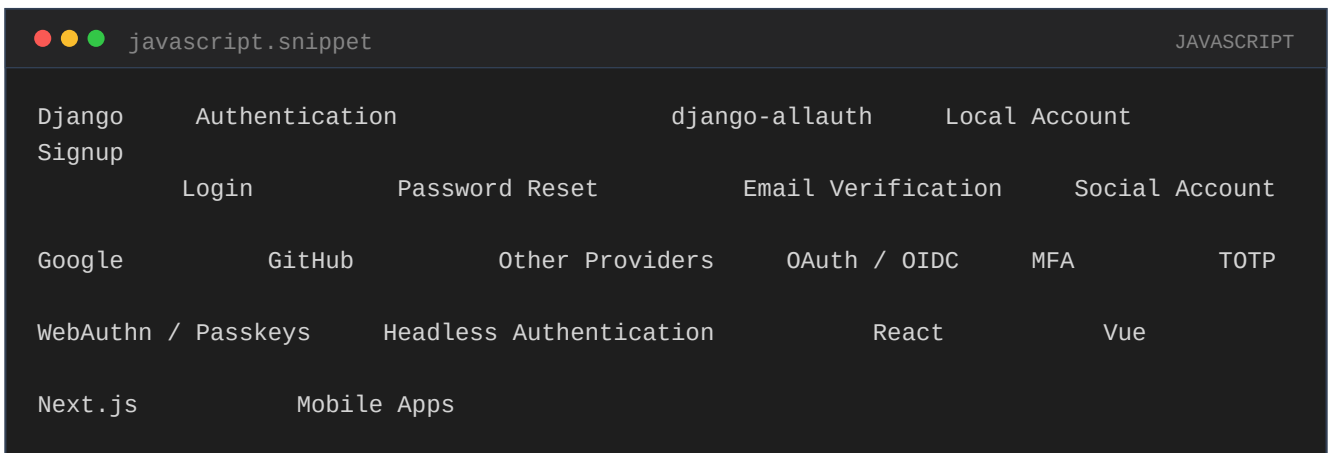
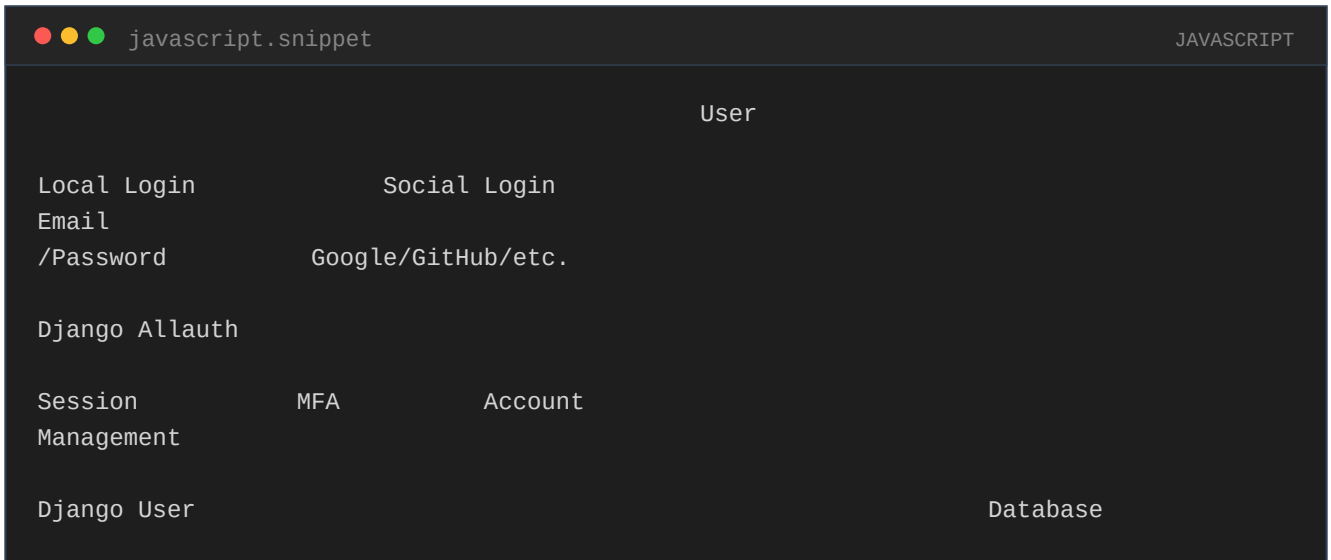
```
javascript.snippet JAVASCRIPT

INSTALLED_APPS = [    "django.contrib.admin",    "django.contrib.auth",
"django.contrib.contenttypes"
,    "django.contrib.sessions",    "django.contrib.messages",
"django.contrib.staticfiles"
,    "django.contrib.sites",    "allauth",    "allauth.account",
"allauth.socialaccount"
,]SITE_ID = = [    "django.contrib.auth.backends.ModelBackend",
"allauth.account.auth_backends.AuthenticationBackend"
,]ACCOUNT_EMAIL_VERIFICATION = "mandatory"ACCOUNT_PREVENT_ENUMERATION =
TrueLOGIN_REDIRECT_URL
= "/dashboard/"LOGOUT_REDIRECT_URL = "/"
```

: Allauth- settings version Allauth releases- settings- configuration project- installed
version- official documentation settings

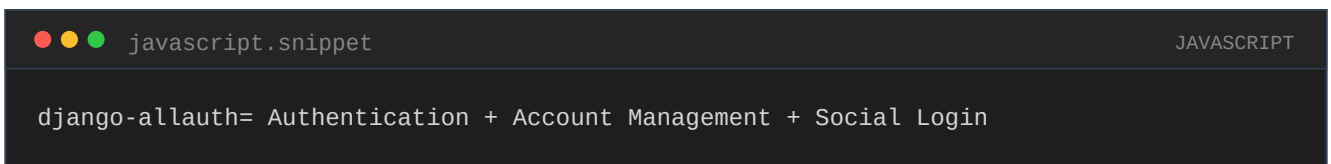
. Authentication Architecture

Django + Allauth architecture:

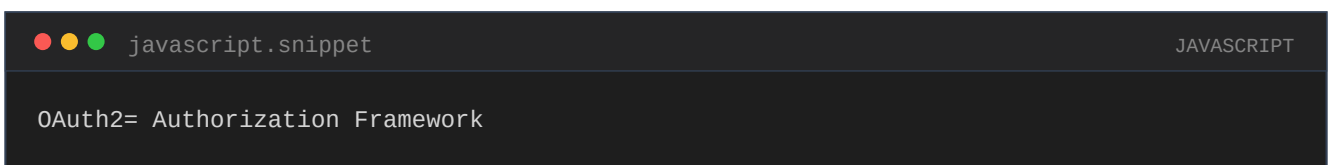


. Quick Revision

Allauth



OAuth2



JWT

```
javascript.snippet JAVASCRIPT

JWT= Token Format
```

Session

```
javascript.snippet JAVASCRIPT

Session= Server-managed authentication state
```

MFA

```
javascript.snippet JAVASCRIPT

MFA= Multiple authentication factors
```

WebAuthn

```
javascript.snippet JAVASCRIPT

WebAuthn= Passwordless / Strong Authentication
```

Adapter

```
javascript.snippet JAVASCRIPT

Adapter= Default Allauth behavior customize mechanism
```

Signal

```
javascript.snippet JAVASCRIPT

Signal= Authentication event- custom action
```

■

```
javascript.snippet JAVASCRIPT

Authentication " ?"Authorization " ?"OAuth2 " resource access permission ?"
JWT
"Claims token format"Session "Server authenticated state ?"MFA
" factor identity verify "
Allauth "Django application- authentication/accountflows- implement framework"
```

. Developer Learning Roadmap

Django developer Allauth sequence:

```
javascript.snippet JAVASCRIPT

1. Django Authentication      2. Custom User Model      3. Signup / Login / Logout
4. Email Verification        5. Password Reset        6. Django Sessions      7. Django
Allauth
8. Social Login              9. OAuth2                 10. OpenID Connect      11. JWT                  12.
DRF Authentication          13. MFA / TOTP            14. WebAuthn / Passkeys 15.
Headless
Authentication              16. Production Security
```

. Final Summary

Django Allauth settings relationship :

```
javascript.snippet JAVASCRIPT

Django Authentication      Allauth      Login
Signup
Email      Password      Verification Reset
Social Authentication      OAuth / OIDC      MFA      WebAuthn /
Passkeys
Headless / API Architecture
```

structure Django- authentication system, social login modern API authentication